TITULACIÓN	MATEMÁTICA COMPUTACIONAL & ING. SOFTWARE	**FECHA**	05/11/2025	U-tad CENTRO UNIVERSITARIO DE TECNOLOGÍA Y ARTE DIGITAL
CURSO	$4^{\underline{o}}$	**HORA**	09:00	
GRUPO	A	**DURACIÓN**	3 HORAS	
ALUMNO	**SOLUCIÓN DEL EXAMEN**			

PROBLEMA 1

Demuestra que $dxdydz = \rho^2 \operatorname{sen}(\varphi) d\theta d\varphi d\rho$ incluyendo todos los pasos intermedios necesarios.

Solución:

$$x = \rho \operatorname{sen}(\varphi)\cos(\theta) \qquad y = \rho \operatorname{sen}(\varphi)\operatorname{sen}(\theta) \qquad z = \rho\cos(\varphi)$$

$$J = \begin{vmatrix} \frac{\partial x}{\partial \rho} & \frac{\partial x}{\partial \varphi} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial \rho} & \frac{\partial y}{\partial \varphi} & \frac{\partial y}{\partial \theta} \\ \frac{\partial z}{\partial \rho} & \frac{\partial z}{\partial \varphi} & \frac{\partial z}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \operatorname{sen}(\varphi)\cos(\theta) & \rho\cos(\varphi)\cos(\theta) & -\rho\operatorname{sen}(\varphi)\operatorname{sen}(\theta) \\ \operatorname{sen}(\varphi)\operatorname{sen}(\theta) & \rho\cos(\varphi)\operatorname{sen}(\theta) & \rho\operatorname{sen}(\varphi)\cos(\theta) \\ \cos(\varphi) & -\rho\operatorname{sen}(\varphi) & 0 \end{vmatrix} =$$

$$= \cos(\varphi) \begin{vmatrix} \rho\cos(\varphi)\cos(\theta) & -\rho\operatorname{sen}(\varphi)\operatorname{sen}(\theta) \\ \rho\cos(\varphi)\operatorname{sen}(\theta) & \rho\operatorname{sen}(\varphi)\cos(\theta) \end{vmatrix}$$

$$- (-\rho\operatorname{sen}(\varphi)) \begin{vmatrix} \operatorname{sen}(\varphi)\cos(\theta) & -\rho\operatorname{sen}(\varphi)\operatorname{sen}(\theta) \\ \operatorname{sen}(\varphi)\operatorname{sen}(\theta) & \rho\operatorname{sen}(\varphi)\cos(\theta) \end{vmatrix} =$$

$$= \cos(\varphi)\left(\rho^2\cos(\varphi)\operatorname{sen}(\varphi)\cos^2(\theta) + \rho^2\cos(\varphi)\operatorname{sen}(\varphi)\operatorname{sin}^2(\theta)\right)$$

$$+ \rho\operatorname{sen}(\varphi)\left(\rho\operatorname{sen}^2(\varphi)\cos^2(\theta) + \rho\operatorname{sen}^2(\varphi)\operatorname{sen}^2(\theta)\right) =$$

$$= \rho^2\cos^2\varphi\operatorname{sen}(\varphi)\left(\cos^2(\theta) + \operatorname{sen}^2(\theta)\right) + \rho^2\operatorname{sen}^3(\varphi)\left(\cos^2(\theta) + \operatorname{sen}^2(\theta)\right) =$$

$$= \rho^2\cos^2(\varphi)\operatorname{sen}(\varphi) + \rho^2\operatorname{sen}^3(\varphi) = \rho^2\operatorname{sen}(\varphi)\left(\cos^2(\varphi) + \operatorname{sen}^2(\varphi)\right) = \rho^2\operatorname{sen}(\varphi)$$

$$\boxed{dxdydz = |J|\, d\theta d\varphi d\theta = \rho^2 \operatorname{sen}(\varphi) d\theta d\varphi d\rho}$$

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PROBLEMA 2

Dado $R = \left\{(x,y) \in \mathbb{R}^2 \mid 1 \leqslant \dfrac{x^2}{4} + \dfrac{y^2}{9} \leqslant 4\right\}$, calcula $\displaystyle\iint_R \frac{|x|}{x^2+y^2}\,dx\,dy$.

Solución:

La región R es simétrica respecto a los ejes X e Y. Por su parte, la función también es simétrica respecto a los dos ejes.

$$f(x,-y) = \frac{|x|}{x^2+(-y)^2} = \frac{|x|}{x^2+y^2} = f(x,y) \qquad f(-x,y) = \frac{|-x|}{(-x)^2+y^2} = \frac{|x|}{x^2+y^2} = f(x,y)$$

Aprovechando esas simetrías, vamos a calcular la integral a partir del resultado en el primer cuadrante haciendo el siguiente cambio de variable:

$$\left.\begin{array}{l} x = 2r\cos(\theta) \\ y = 3r\,\mathrm{sen}(\theta) \end{array}\right\} \Longrightarrow J = \begin{vmatrix} \dfrac{\partial x}{\partial r} & \dfrac{\partial x}{\partial \theta} \\ \dfrac{\partial y}{\partial r} & \dfrac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} 2\cos(\theta) & -2r\,\mathrm{sen}(\theta) \\ 3\,\mathrm{sen}(\theta) & 3r\cos(\theta) \end{vmatrix} = 6r$$

$$I = 4\iint_{R_1} \frac{x}{x^2+y^2}\,dx\,dy = 4\int_{\theta=0}^{\theta=\frac{\pi}{2}} \left(\int_{r=1}^{r=2} \frac{2r\cos(\theta)}{4r^2\cos^2(\theta)+9r^2\,\mathrm{sen}^2(\theta)}\,6r\,dr\right) d\theta =$$

$$= 48\int_{\theta=0}^{\theta=\frac{\pi}{2}} \left(\int_{r=1}^{r=2} \frac{\cancel{r^2}\cos(\theta)}{\cancel{r^2}\,(4\cos^2(\theta)+9\,\mathrm{sen}^2(\theta))}\,dr\right) d\theta = 48\int_{\theta=0}^{0=\frac{\pi}{2}} \left(\int_{r=1}^{r=2} \frac{\cos(\theta)}{4+5\,\mathrm{sen}^2(\theta)}\,dr\right) d\theta =$$

$$= 48\int_{\theta=0}^{\theta=\frac{\pi}{2}} \frac{\cos(\theta)}{4+5\,\mathrm{sen}^2(\theta)}\Big[r\Big]_{r=1}^{r=2} d\theta = 48\int_{\theta=0}^{\theta=\frac{\pi}{2}} \frac{\cos(\theta)}{4+5\,\mathrm{sen}^2(\theta)}\,d\theta =$$

$$= \left\{\begin{array}{l} t = \mathrm{sen}(\theta) \longrightarrow dt = \cos(\theta)d\theta \\ \theta = 0 \longrightarrow t = 0, \theta = \dfrac{\pi}{2} \longrightarrow t = 1 \end{array}\right\} = 48\int_0^1 \frac{dt}{4+5t^2} = \frac{48}{4}\int_0^1 \frac{1}{1+\frac{5}{4}t^2}\,dt =$$

$$= 12\int_0^1 \frac{1}{1+\left(\frac{\sqrt{5}}{2}t\right)^2}\,dt = 12\frac{2}{\sqrt{5}}\int_0^1 \frac{\frac{\sqrt{5}}{2}}{1+\left(\frac{\sqrt{5}}{2}t\right)^2}\,dt =$$

$$= \frac{24}{\sqrt{5}}\left[\arctan\left(\frac{\sqrt{5}}{2}t\right)\right]_0^1 = \frac{24}{\sqrt{5}}\left(\arctan\left(\frac{\sqrt{5}}{2}\right) - \arctan(0)\right) =$$

$$= \boxed{\frac{24}{\sqrt{5}}\arctan\left(\frac{\sqrt{5}}{2}\right) \approx 9.027}$$

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PROBLEMA 3

Calcula la integral $\iint_R x^2 \cos(xy)\,\text{sen}\left(\frac{x}{y}\right) dxdy$, donde el recinto de integración es R:

$$R = \left\{(x,y) \in \mathbb{R}^2 \;\middle|\; \frac{\pi}{2x} \leqslant y \leqslant \frac{\pi}{x},\ \frac{\pi y}{2} \leqslant x \leqslant \pi y,\ x>0,\ y>0\right\}$$

Solución:

Vamos a realizar un cambio de variables:

$$\left.\begin{array}{l} u = xy \\ v = \dfrac{x}{y} \end{array}\right\} \Longrightarrow J = \frac{1}{\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}} = \frac{1}{\begin{vmatrix} y & x \\ \frac{1}{y} & -\frac{x}{y^2} \end{vmatrix}} = \frac{1}{-2\cdot\frac{x}{y}} = -\frac{1}{2v}$$

Puesto que $x>0$ e $y>0$, el valor absoluto del Jacobiano es $|J| = \frac{1}{2}\frac{y}{x} = \frac{1}{2v}$.

$$\iint_R x^2 \cos(xy)\,\text{sen}\left(\frac{x}{y}\right) dxdy = \int_{u=\frac{\pi}{2}}^{u=\pi}\int_{v=\frac{\pi}{2}}^{v=\pi} u\cancel{v}\cos(u)\,\text{sen}(v)\frac{1}{2\cancel{v}}dvdu =$$

$$= \frac{1}{2}\left(\int_{u=\frac{\pi}{2}}^{u=\pi} u\cos(u)du\right)\left(\int_{v=\frac{\pi}{2}}^{v=\pi}\text{sen}(v)dv\right) = \left\{\begin{array}{lll} U = u & \longrightarrow & dU = du \\ dV = \cos(u)du & \longrightarrow & V = \text{sen}(u) \end{array}\right\} =$$

$$= \frac{1}{2}\left(\Big[u\,\text{sen}(u)\Big]_{\frac{\pi}{2}}^{\pi} - \int_{\frac{\pi}{2}}^{\pi}\text{sen}(u)du\right)\Big[-\cos(v)\Big]_{\frac{\pi}{2}}^{\pi} =$$

$$= \frac{1}{2}\Big[u\,\text{sen}(u)+\cos(u)\Big]_{\frac{\pi}{2}}^{\pi}\Big[-\cos(v)\Big]_{\frac{\pi}{2}}^{\pi} = \frac{1}{2}\left(-1-\frac{\pi}{2}\right)\big(-(-1)\big) = \boxed{-\frac{2+\pi}{4} \approx -1.28}$$

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PROBLEMA 4

Calcula la integral impropia $\iiint_D \dfrac{1+z^2}{\sqrt{x^2+y^2+z^2}}dV$, donde la región de integración es:

$$D=\{(x,y,z)\in\mathbb{R}^3 \mid z^2\leqslant 3(x^2+y^2),\ 3z^2\geqslant x^2+y^2,\ x^2+y^2+z^2\leqslant 4,\ z\geqslant 0\}$$

Solución:

Vamos a utilizar coordenadas esféricas en este problema, teniendo en cuenta que el valor de φ se encuentra en el rango $\left[\dfrac{\pi}{6},\dfrac{\pi}{3}\right]$.

$$\iiint_D \frac{1+z^2}{\sqrt{x^2+y^2+z^2}}dV=\lim_{\epsilon\to 0^+}\int_{\theta=0}^{\theta=2\pi}\left(\int_{\varphi=\frac{\pi}{6}}^{\varphi=\frac{\pi}{3}}\left(\int_{\rho=\epsilon}^{\rho=2}\frac{1+\rho^2\cos^2(\varphi)}{\rho}\rho^2\operatorname{sen}(\varphi)d\rho\right)d\varphi\right)d\theta=$$

$$=\lim_{\epsilon\to 0^+}\int_{\theta=0}^{\theta=2\pi}\left(\int_{\varphi=\frac{\pi}{6}}^{\varphi=\frac{\pi}{3}}\left(\int_{\rho=\epsilon}^{\rho=2}\left(1+\rho^2\cos^2(\varphi)\right)\rho\operatorname{sen}(\varphi)d\rho\right)d\varphi\right)d\theta=$$

$$=\lim_{\epsilon\to 0^+}\int_{\theta=0}^{\theta=2\pi}\left(\int_{\varphi=\frac{\pi}{6}}^{\varphi=\frac{\pi}{3}}\left(\int_{\rho=\epsilon}^{\rho=2}\left(\rho\operatorname{sen}(\varphi)+\rho^3\operatorname{sen}(\varphi)\cos^2(\varphi)\right)d\rho\right)d\varphi\right)d\theta=$$

$$=\lim_{\epsilon\to 0^+}\int_{\theta=0}^{\theta=2\pi}\left(\int_{\varphi=\frac{\pi}{6}}^{\varphi=\frac{\pi}{3}}\left[\frac{\rho^2}{2}\operatorname{sen}(\varphi)+\frac{\rho^4}{4}\operatorname{sen}(\varphi)\cos^2(\varphi)\right]_{\rho=\epsilon}^{\rho=2}d\varphi\right)d\theta=$$

$$=\lim_{\epsilon\to 0^+}\int_{\theta=0}^{\theta=2\pi}\left(\int_{\varphi=\frac{\pi}{6}}^{\varphi=\frac{\pi}{3}}\left(2\operatorname{sen}(\varphi)+4\operatorname{sen}(\varphi)\cos^2(\varphi)-\frac{\epsilon^2}{2}\operatorname{sen}(\varphi)-\frac{\epsilon^4}{4}\operatorname{sen}(\varphi)\cos^2(\varphi)\right)d\varphi\right)d\theta=$$

$$=\left(\int_{\theta=0}^{\theta=2\pi}d\theta\right)\left(\int_{\varphi=\frac{\pi}{6}}^{\varphi=\frac{\pi}{3}}\left(2\operatorname{sen}(\varphi)+4\operatorname{sen}(\varphi)\cos^2(\varphi)\right)d\varphi\right)=$$

$$=\left[\theta\right]_{\theta=0}^{\theta=2\pi}\left[-2\cos(\varphi)-\frac{4}{3}\cos^3(\varphi)\right]_{\varphi=\frac{\pi}{6}}^{\varphi=\frac{\pi}{3}}=$$

$$=2\pi\left(\left(-2\cos\left(\frac{\pi}{3}\right)-\frac{4}{3}\cos^3\left(\frac{\pi}{3}\right)\right)-\left(-2\cos\left(\frac{\pi}{6}\right)-\frac{4}{3}\cos^3\left(\frac{\pi}{6}\right)\right)\right)=$$

$$=2\pi\left(\left(-2\cdot\frac{1}{2}-\frac{4}{3}\cdot\frac{1}{8}\right)-\left(-2\frac{\sqrt{3}}{2}-\frac{4}{3}\frac{3\sqrt{3}}{8}\right)\right)=$$

$$=2\pi\left(-\frac{7}{6}+\frac{3\sqrt{3}}{2}\right)=2\pi\frac{9\sqrt{3}-7}{6}=\boxed{\frac{\pi}{3}\left(9\sqrt{3}-7\right)\approx 8.99}$$